

Assignment - 4.

Power rings: These are wide metal loops (V_{DD} & V_{SS}) that run around the periphery of a chip, a macro or a block.

Purpose: act as a stable low resistance boundary that collects current from the top-level power pads/pads ring and distributes it into the core or macro. provides uniform access point.

Power Straps: these are wide regularly spaced parallel metal wires (alternating V_{DD}/V_{SS}) that run vertically and/or horizontally.

Purpose: distribute power from chip to interior area of chip.

Act as mesh that provides redundant current paths feed the lower level power rails via stacked vias at regular intervals.

Power Rails: are the lowest level, narrowest power/ground metal lines. They run horizontally through every standard cell.

Purpose: provide the final direct connection.

General Hierarchy

I/O pads → power ring



Power Strap



Power rail



Standard cells